

PAPER_709: Westerlund 2 Young Super Star Cluster: UQFF MUGE

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Abstract

Westerlund 2 (Wd 2) is one of the youngest and most massive star clusters in the Milky Way, located at 15,000 ly in the Carina-WR constellation. Its mass ($\sim 3 \times 10^4 M_\odot$) and age (~ 2 Myr) make it an ideal test case for rapid gas-to-star conversion dynamics. The UQFF MUGE is derived with the dynamic mass growth function incorporating the Wolf-Rayet star contribution.

1. System Parameters

Parameter	Value	Description
M_{init}	$3 \times 10^4 M_\odot$	Initial stellar mass
r_{wd2}	9.461×10^{16} m	Cluster radius (10 ly)
M_0^{gas}	3.333	Gas-to-star mass ratio
τ_{SF}	6.312×10^{13} s	SF timescale (2 Myr)
B_{wd2}	10^{-5} T	Cluster magnetic field

2. Master UQFF Gravity Equation

$$g_{W2}(t) = \frac{GM(t)}{r^2} (1 + H_0 t) \left(1 - \frac{B}{B_{crit}} \right) + (U_{g1} + U_{g4})(1 + f_{TRZ}) + \frac{\Lambda c^2}{3} + q(v \times B) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}} \right) \times 10^{-}$$

2.1 Rapid Gas Conversion (Wolf-Rayet + OB mass infall)

$$M(t) = M_{init} (1 + 3.333 e^{-t/\tau_{SF}})$$

At $t = 0$: $M = 4.333 M_{init}$; at $t = \tau_{SF}$: $M = 2.226 M_{init}$.

3. Numerical Result

$$g_{W2}(t = 1 \text{ Myr}) \approx 1.053 \times 10^{-3} \text{ m/s}^2$$

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$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Clarkson et al. (2012), ApJ, 751, 132
- Zeidler et al. (2015), A&A, 576, A28
- UQFF Framework, Star-Magic Session 175

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